Video 11 - Overfitting

Video Contents

Video Synopsis

Introduction ***05:00 – 13:25***

Overfitting is introduced as:

1. Choosing a model that is too complex, thus allowing overfitting because the model is so ‘wiggly’ that it captures the noise in your data (example of polynomials).
2. Allowing a model to learn too long on the data, until the point that Eout goes up again whilst Ein still goes down (the neural net example)

Noise causes overfitting ***13:25 – 15:40***

Noise is introduced as the source of overfitting: if you allow it, your model will try to perfectly fit every data point, including the noise superimposed on each data point. In other words, it will treat noise as part of the target value/function.

Stochastic noise and Deterministic noise ***15:40-22:14***

Using an example, the two types of noise are introduced. The first (for the low order target) is good old stochastic noise. The second one is deterministic noise (though this term is not introduced until later).

The amount of data will determine what models you can choose ***22:14 – 24:55***

Prof Mostafa links back to our knowledge of the Theory of Generalisation: even if you know that data was generated with, say, a 10^{th} order polynomial, the amount of data you have, may mean that you cannot use a 10^{th} order model to fit the data because you don’t have enough data to fulfil the requirements imposed by the theory of generalization ($N>= 10*d_{VC}$). In the video, an example is given where a second order model yields far better results than a 10^{th} order model on data that was generated using a 10^{th} order polynomial.

Overfitting and the Learning Curve ***24:55 – 27:40***

Overfitting is indicated as the left-hand side of the learning curve (before Ein and Eout have converged) and this shows that overfitting is strongly related to the amount of data available for training: the more training data, the lower the “over-fit” will be.

SAFE SKIP! Proving the relationship between overfitting, noise and complexity ***27:40 – 42:39***

Example introduced with some maths that shows that overfitting increases with increasing noise, but also with increasing complexity of the target function.

Definition of Deterministic Noise ***42:39 – 47:20***

Deterministic noise is introduced as the part of the target function that your hypothesis set cannot capture (due to limited complexity).

<u>SAFE SKIP!</u> Why does deterministic noise lead to overfitting? ***47:20 – 51:12***

The reason why deterministic noise leads to overfitting is elaborated on. The reason is that a finite data set will allow some fitting of the deterministic noise that does not translate to out-of-sample.

<u>SAFE SKIP!</u> The relationship between deterministic noise and bias in the bias-variance analysis ***51:12-58:16***

It is shown that deterministic noise is the same as the bias component in the bias-variance trade-off.

Introduction to Regularization ***58:16 - 1:01:55***

The two cures to overfitting are regularization and validation. Regularization is introduced as 'putting on the brakes' during the learning process. Validation is introduced as a check after training to see how well the model performs out of sample.
In an example Prof. Mostafa shows that the impact of regularization, even if applied in a tiny dose, can be very beneficial.

<u>SAFE SKIP!</u> Q&A ***1:01:55 – end***